finaldelim

Better-sized delimiters, readable nesting, simpler line breaks

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A formula may start as a short expression, acquire a tall fraction and nested brackets, and eventually need several lines. finaldelim lets you keep writing ordinary `\left`, `\middle` and `\right` as those needs change. This manual follows that progression: adjust the size, make the nesting visible, then break and align the formula.

Compile with **LuaLaTeX**. To start, add:

```
\usepackage[render=draw]{finaldelim}
```

Weighted sizing is on by default; the example above also enables continuous drawing. Colors remain opt-in. The examples use Latin Modern Math, selected with `unicode-math`; The manual uses `render=draw`, except in native-rendering comparisons. finaldelim does not select a font. It loads `amsmath` and `xcolor`.

1 A tall term makes the brackets look too large

Suppose a bracketed expression has three low terms, a tall central fraction, and three more low terms: the profile `lllhlll`. Native sizing sees the tallest content and gives both brackets enough height for it, even though their immediate neighbours are small.

The following schematic expression isolates that problem. The same source is actually typeset twice, with ordinary mathematical spacing and the same font. The native result uses native sizing and font glyphs; the finaldelim result uses the default weighted sizing with continuous drawing.

One source for both results

```
\begin{equation*}
 \left[a+b+c+
 \frac{\displaystyle\sum_{i=1}^n
              \frac{x_i^2}{1+x_i^2}}
      {\displaystyle\sum_{j=1}^m
              \frac{y_j^2}{1+y_j^2}}
 +d+e+f\right]
\end{equation*}
```

Native sizing

$$\left[a+b+c+\frac{\displaystyle\sum_{i=1}^{n}\frac{x_i^2}{1+x_i^2}}{\displaystyle\sum_{j=1}^{m}\frac{y_j^2}{1+y_j^2}}+d+e+f\right]$$

Weighted sizing, continuous drawing

$$\left[a+b+c+\frac{\displaystyle\sum_{i=1}^{n}\frac{x_i^2}{1+x_i^2}}{\displaystyle\sum_{j=1}^{m}\frac{y_j^2}{1+y_j^2}}+d+e+f\right]$$

The weighted result gives nearby content more influence. The brackets form a frame around the expression without needing to enclose the full height of the distant fraction. Both ends still receive a shared size target. No manual `\bigl` or `\bigr` commands are needed.

The rest of this section explains the size controls and then the optional continuous renderer. For nesting colors, go to Section 2; for line breaks, go to Section 3.

1.1 Make the frame more or less compact

The defaults are `lambda=20pt`, `q=3`, and `scale=1`. Usually no adjustment is needed. To ask for a fuller frame, increase `scale`; to give distant content less influence, decrease `lambda`.

```
\FinalDelimSetup{scale=1.1}     % fuller frame in subsequent formulas
\begin{equation}[scale=0.9]    % this display only
  \left[a+\frac{b}{c}+d\right]
\end{equation}
```

The default scale uses the full estimated radius. Setting `scale=0.9` reduces it by ten percent about the math axis. It changes the target size, not the shape of a glyph. With native font rendering, several nearby targets can select the same prebuilt glyph. `sizing=native` restores the native size policy; colors and line breaking remain available.

1.2 Why distance, width, and height matter

The purpose of the model is to balance a delimiter against the content near it while retaining the influence of taller material farther away. Treat the formula as a horizontal profile of boxes. For box i, let h_i and d_i be its height and depth, and let a be the math axis height. Its radius is

$$r_i = \max(h_i - a,\, d_i + a).$$

This radius applies over the box's actual width: a broad tall term contributes more than a narrow term of the same height. Spaces take the smallest measured radius in the group. Write the resulting profile as $r(x)$ over width W. A delimiter at position s_j uses

$$w_j(x) = e^{-|x-s_j|/\lambda}, \qquad R_j = \left(\frac{\int_0^W r(x)^q w_j(x)\,dx}{\int_0^W w_j(x)\,dx} \right)^{1/q}.$$

At a distance of `lambda`, a contribution has $1/e$ of its weight at the delimiter. The fixed default of 20pt is a physical length, independent of font size. The power q gives tall material extra influence even when it occupies only a small part of the width.

The shared target radius is

$$R = \texttt{scale}\ \max_j R_j.$$

For a pair, the delimiter positions are 0 and W. Taking the maximum keeps the estimate at an end close to tall content from being diluted by the other end. Every `\middle` position also contributes when present. Nested groups are sized independently.

The renderer turns this target into a visible delimiter. The default renderer selects a font glyph or assembly. The next page shows how to obtain continuous intermediate sizes and clean joins instead.

1.3 The size is right, but the outline has steps or joins

A font supplies a finite set of prebuilt delimiter sizes and may assemble larger ones from repeated pieces. To interpolate between those sizes and paint assembled outlines without repeated painting at the joins, enable continuous drawing:

```
\usepackage[render=draw]{finaldelim}
```

Continue using the same delimiter commands. The renderer derives its shapes from the current math font. It needs no additional package or directory. The visible delimiter height is exactly $2R$, with the target radius R defined above; `scale` is applied only once. Font sizes serve as shape samples, not size steps. Between samples the renderer interpolates; beyond them it extends the font's construction to the requested length. The delimiter boxes and attached scripts use that same final extent.

Here is a Frobenius norm with a tall matrix.[1] The sizing policy is identical in both results; only rendering changes. **Font glyphs: `render=native`**

LaTeX source

```
\begin{equation*}
 \left\Vert\begin{matrix}
 a_{11}&a_{12}\\ a_{21}&a_{22}\\
 a_{31}&a_{32}\\ a_{41}&a_{42}
 \end{matrix}\right\Vert_F
 =\sqrt{\sum_{i=1}^4\sum_{j=1}^2
              |a_{ij}|^2}
\end{equation*}
```

Result

$$\left\Vert\begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{matrix}\right\Vert_F = \sqrt{\sum_{i=1}^{4}\sum_{j=1}^{2} |a_{ij}|^2}$$

Continuous outlines: `render=draw`

LaTeX source

```
\begin{equation*}
 \left\Vert\begin{matrix}
 a_{11}&a_{12}\\ a_{21}&a_{22}\\
 a_{31}&a_{32}\\ a_{41}&a_{42}
 \end{matrix}\right\Vert_F
 =\sqrt{\sum_{i=1}^4\sum_{j=1}^2
              |a_{ij}|^2}
\end{equation*}
```

Result

$$\left\Vert\begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{matrix}\right\Vert_F = \sqrt{\sum_{i=1}^{4}\sum_{j=1}^{2} |a_{ij}|^2}$$

Change the renderer for subsequent mathematics with `\FinalDelimSetup{render=draw}`, or for one display with `\begin{equation}[render=draw]`. Use `render=native` to return to font glyphs. Rendering and sizing are independent choices.

Drawing also works with niji and line breaks. Scripts after a visible closing delimiter follow its final size; the formula contents retain their full height and depth for surrounding layout. Unsupported fonts or shapes retain native rendering. Drawn delimiters are PDF paths and do not provide text extraction or accessibility tagging.

[1] Wikipedia, Matrix norm, Frobenius norm; adapted to a four-by-two matrix.

See the joins in a rasterized view

At screen resolution, overlapping assembly pieces can leave regularly spaced dark bands. The following is a real rasterization of a 64pt double bar in Latin Modern Math. Both versions use the same font, target size and rasterizer. The only change is `render=native` versus `render=draw`.

Native assembly **Continuous outline**



Look at the repeated horizontal bands in the left-hand bars. They occur at the joins and give the strokes a segmented appearance. The right-hand bars are painted as a single filled outline, without those bands.

These images were rendered from PDF by Poppler at 96 dpi and enlarged with nearest-neighbour sampling so individual pixels remain visible. No bands have been drawn in, removed or enhanced. Their appearance in a PDF viewer depends on resolution and rasterization; this comparison records one reproducible view. The source and generation script are included in `manual-assets/`.

Keep native root geometry, but draw clean joins

The same `render=draw` setting also applies to `\sqrt` and `\sqrt[n]`. TeX first lays out the root normally. finaldelim then paints the selected root sign and its overbar together as one filled outline.

Native rendering

LaTeX source

```
\begin{equation*}
 \sqrt[3]{
   \frac{\displaystyle\sum_{i=1}^n x_i^2}
        {\displaystyle\sum_{j=1}^m y_j^2}
 }^{u}_{v}
\end{equation*}
```

Result

$$\sqrt[3]{\frac{\displaystyle\sum_{i=1}^n x_i^2}{\displaystyle\sum_{j=1}^m y_j^2}}^{u}_{v}$$

Continuous drawing

LaTeX source

```
\begin{equation*}
 \sqrt[3]{
   \frac{\displaystyle\sum_{i=1}^n x_i^2}
        {\displaystyle\sum_{j=1}^m y_j^2}
 }^{u}_{v}
\end{equation*}
```

Result

$$\sqrt[3]{\frac{\displaystyle\sum_{i=1}^n x_i^2}{\displaystyle\sum_{j=1}^m y_j^2}}^{u}_{v}$$

The root index, superscript, subscript, radicand and surrounding spacing keep their native positions. The weighted delimiter-size policy is not applied to the root: its bar still clears the full height of the radicand. Nested roots and inline roots use the same drawing path, with no extra syntax.

Unsupported font outlines retain native rendering. As with drawn delimiters, the root sign becomes a PDF path rather than extractable text. Integral signs retain their native glyphs and limit positioning; ordinary integrals in the tested fonts use complete glyphs rather than stacked extender pieces.

Long braces, arrows and accents

Long horizontal symbols can have the same repeated joins as tall delimiters. With `render=draw`, finaldelim paints horizontal font assemblies and amsmath's repeated arrow pieces as a single outline. Continue using the usual commands:

Purpose	Commands
Group terms above or below	`\overbrace`, `\underbrace`
Label a long arrow	`\xrightarrow`, `\xleftarrow`
Put an arrow over an expression	`\overrightarrow`, `\overleftarrow`, `\overleftrightarrow`
Use a wide accent	`\widehat`, `\widetilde`

Native rendering

LaTeX source

```
\begin{equation*}
 \underbrace{x_1+\cdots+x_n}
            _{n\text{ terms}}\\
 A\xrightarrow[n\to\infty]
              {\text{convergence}}B\\
 \overrightarrow{AB}
 +\widehat{xyz}+\widetilde{xyz}
\end{equation*}
```

Result

$$\begin{gathered}\underbrace{x_1+\cdots+x_n}_{n\text{ terms}}\\ A\xrightarrow[n\to\infty]{\text{convergence}}B\\ \overrightarrow{AB}+\widehat{xyz}+\widetilde{xyz}\end{gathered}$$

Continuous drawing

LaTeX source

```
\begin{equation*}
 \underbrace{x_1+\cdots+x_n}
            _{n\text{ terms}}\\
 A\xrightarrow[n\to\infty]
              {\text{convergence}}B\\
 \overrightarrow{AB}
 +\widehat{xyz}+\widetilde{xyz}
\end{equation*}
```

Result

$$\begin{gathered}\underbrace{x_1+\cdots+x_n}_{n\text{ terms}}\\ A\xrightarrow[n\to\infty]{\text{convergence}}B\\ \overrightarrow{AB}+\widehat{xyz}+\widetilde{xyz}\end{gathered}$$

Lengths, attachment points, text labels and scripts keep their native positions. For a wide accent supplied as a complete glyph, drawing preserves that glyph's native shape and size; it does not stretch it to cover more of the expression. Ordinary fraction rules, overlines and underlines remain native rules.

Unsupported outline constructions retain native rendering. Like the other drawn symbols, these outlines become PDF paths rather than extractable glyphs.

2 Nested brackets become hard to follow

Once a formula contains several levels of parentheses, color can make the matching pairs easier to find. Add `niji`:

```
\usepackage[niji]{finaldelim}
% Combine with continuous drawing if desired:
% \usepackage[niji,render=draw]{finaldelim}
```

Each nesting level gets a color. The contents keep their usual color; ordinary parentheses written without `\left` and `\right` are not scanned. `rainbow` is an alias for `niji`.

The normal cumulative distribution function contains an inner argument of the error function.[2] **Without niji**

LaTeX source

```
\begin{equation*}[niji=false]
 F(x)=\frac12\left[
 1+\operatorname{erf}\left(
 \frac{x-\mu}{\sigma\sqrt2}
 \right)\right].
\end{equation*}
```

Result

$$F(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right].$$

With niji

LaTeX source

```
\begin{equation*}[niji]
 F(x)=\frac12\left[
 1+\operatorname{erf}\left(
 \frac{x-\mu}{\sigma\sqrt2}
 \right)\right].
\end{equation*}
```

Result

$$F(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right].$$

2.1 Use the same colors inline

No special inline command is needed.

LaTeX source

```
\FinalDelimSetup{niji}
The standardized argument is
$\left(\frac{x-\mu}{\sigma}
\right)$.
```

Result

The standardized argument is $\left(\frac{x-\mu}{\sigma}\right)$.

2.2 Choose a palette or turn it off locally

```
\FinalDelimSetup{niji,colors={blue,red,teal}}
\begin{equation}[niji=false]
  \left[a+\left(b+c\right)\right]
\end{equation}
```

The palette repeats at deeper levels. The environment option affects only that display; the surrounding niji setting resumes afterwards.

[2] Wikipedia, Normal distribution, cumulative distribution function.

2.3 A delimiter is also needed inside the pair

Use `\middle` without ending and restarting the surrounding group. There may be more than one middle delimiter. A matrix element in bra–ket notation supplies a simple example.[3]

LaTeX source

```
\begin{equation*}[niji]
 \left\langle\phi
 \middle|\hat A\middle|\psi
 \right\rangle
\end{equation*}
```

Result

$$\left\langle\phi\middle|\hat A\middle|\psi\right\rangle$$

Both bars use the color of the surrounding pair. All four positions contribute to the shared size target described earlier; a middle delimiter close to tall content can therefore increase that target. The visible extents also depend on the particular delimiter shapes supplied by the font.

3 The formula no longer fits on one line

Choose the break with `\\` and the alignment point with `&`. Keep one delimiter pair even when `\left` and `\right` occur on different rows. In the default mode, write directly inside `equation`.

3.1 Remove the extra scaffolding from a long formula

The divergence between two multivariate normal distributions contains a trace, a quadratic form and a logarithmic determinant.[4] It is long enough that putting these terms on separate rows helps the reader. The next page compares the same break points in plain amsmath and finaldelim.

What the traditional form needs

- An inner `split` environment.
- `\right.` on the opening row and `\left.` on the closing row.
- A `\vphantom` to give both brackets the same size when their rows have different heights.

The intermediate row needs no invisible delimiter pair. The comparison marks only the additional code in **red bold**.

[3] Wikipedia, Bra–ket notation, linear operators acting on bras.
[4] Wikipedia, Kullback–Leibler divergence, multivariate normal distributions.

Plain amsmath

LaTeX source

```
\begin{equation}
 \begin{split}
 &D_{\mathrm{KL}}(\mathcal N_0
     \Vert\mathcal N_1)\\
 &=\frac12\left[
   \vphantom{\ln\frac{\det\Sigma_1}
                     {\det\Sigma_0}}
   \operatorname{tr}(
     \Sigma_1^{-1}\Sigma_0)-k\right.\\
 &\quad+(\mu_1-\mu_0)^{\mathsf T}
   \Sigma_1^{-1}(\mu_1-\mu_0)\\
 &\quad\left.+\ln\frac{\det\Sigma_1}
                  {\det\Sigma_0}
   \right].
 \end{split}
\end{equation}
```

Result

$$
\begin{aligned}
&D_{\mathrm{KL}}(\mathcal N_0\Vert\mathcal N_1)\\
&=\frac12\left[\operatorname{tr}(\Sigma_1^{-1}\Sigma_0)-k\right.\\
&\quad+(\mu_1-\mu_0)^{\mathsf T}\Sigma_1^{-1}(\mu_1-\mu_0)\\
&\quad\left.+\ln\frac{\det\Sigma_1}{\det\Sigma_0}\right].
\end{aligned}
\tag{1}
$$

finaldelim: direct breaks, with niji

LaTeX source

```
\begin{equation}[niji]
 &D_{\mathrm{KL}}(\mathcal N_0
     \Vert\mathcal N_1)\\
 &=\frac12\left[
   \operatorname{tr}\left(
     \Sigma_1^{-1}\Sigma_0\right)-k\\
 &\quad+(\mu_1-\mu_0)^{\mathsf T}
   \Sigma_1^{-1}(\mu_1-\mu_0)\\
 &\quad+\ln\frac{\det\Sigma_1}
                  {\det\Sigma_0}
   \right].
\end{equation}
```

Result

$$
\begin{aligned}
&D_{\mathrm{KL}}(\mathcal N_0\Vert\mathcal N_1)\\
&=\frac12\left[\operatorname{tr}\left(\Sigma_1^{-1}\Sigma_0\right)-k\right.\\
&\quad+(\mu_1-\mu_0)^{\mathsf T}\Sigma_1^{-1}(\mu_1-\mu_0)\\
&\quad\left.+\ln\frac{\det\Sigma_1}{\det\Sigma_0}\right].
\end{aligned}
\tag{2}
$$

The finaldelim source keeps a single `\left`/`\right` pair and the chosen alignment points. There is no inner alignment wrapper or height phantom. Adding `niji` colors the nested pairs, including the pair that crosses rows. Both versions give the whole expression one equation number.

3.2 Choose what determines a spanning pair's size

The default `spanning=folded` lets tall middle content influence the brackets without giving it its full horizontal extent in the measurement. Use `spanning=endpoints` to size from the delimiter-bearing rows alone, or `spanning=full` to weight the whole continuous expression.

Here the earlier `lllhlll` expression is broken into three rows. The contents and break points are identical; only the spanning policy changes. Use this source, changing only the `spanning` option:

```
\begin{equation*}[spanning=folded]
 F&=\left[a+b+c\\
  &\quad +\frac{\displaystyle\sum_{i=1}^n
              \frac{x_i^2}{1+x_i^2}}
      {\displaystyle\sum_{j=1}^m
              \frac{y_j^2}{1+y_j^2}}\\
  &\quad +d+e+f\right]
\end{equation*}
```

endpoints

$$F = [a + b + c + \frac{\displaystyle\sum_{i=1}^{n} \frac{x_i^2}{1+x_i^2}}{\displaystyle\sum_{j=1}^{m} \frac{y_j^2}{1+y_j^2}} + d + e + f]$$

folded (default)

$$F = \big[a + b + c + \frac{\displaystyle\sum_{i=1}^{n} \frac{x_i^2}{1+x_i^2}}{\displaystyle\sum_{j=1}^{m} \frac{y_j^2}{1+y_j^2}} + d + e + f\big]$$

full

$$F = \Big[a + b + c + \frac{\displaystyle\sum_{i=1}^{n} \frac{x_i^2}{1+x_i^2}}{\displaystyle\sum_{j=1}^{m} \frac{y_j^2}{1+y_j^2}} + d + e + f\Big]$$

The endpoint result follows the low terms beside the brackets. The folded result also accounts for the tall middle content, represented for measurement by a box of width `fold-width` (default 0.1em), retaining its maximum height and depth. The full result uses the actual widths of all the contents. Choose the framing that suits the formula. Folding can also bring tall content closer to a delimiter, so its result is not necessarily smaller than `full`.

`endpoints` uses the maximum height and depth of the first and last rows inside the pair, plus any rows containing its `\middle` delimiters. `folded` retains the horizontal profiles of delimiter-bearing rows and replaces each intervening run of rows by one representative box. It applies the same distance weighting and never uses a target smaller than `endpoints`. The actual contents and line breaks are unchanged. Set the representative width with `fold-width`, for example:

```
\FinalDelimSetup{fold-width=0.1em}
\begin{equation}[fold-width=0.5em]
  ...
\end{equation}
```

The option also works at package level. Positive TeX lengths such as `0.01em` or `0.1pt` are accepted; font-relative units are evaluated when the formula is typeset. `full` applies the distance-weighted model to the continuous contents. Pairs contained on one row use weighted sizing in either mode. `\\[3pt]` adds space after a row; `\jot` controls the usual extra row spacing. Row layout and numbering follow amsmath conventions.

3.3 Give each row its own number

Use `align` when the rows need separate numbers. The estimates of a normal distribution's mean and variance make a short example.[5]

LaTeX source

```
\begin{align}
 \hat\mu
 &=\frac1n\sum_{i=1}^n x_i
   \label{eq:mean}\\
 \hat\sigma^2
 &=\frac1n\sum_{i=1}^n
   (x_i-\hat\mu)^2
   \label{eq:variance}
\end{align}
```

Result

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{3}$$

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{\mu})^2 \tag{4}$$

The labels refer separately to the mean, (3), and the variance, (4). Use `\notag` to suppress a row's number, or `align*` to suppress all automatic numbers. `equation*` and `\[...\]` give an unnumbered display with one or more rows.

3.4 Align more than one pair of columns

Additional `&` markers create additional alignment pairs. Use `alignat` to specify the number of pairs and arrange their spacing explicitly. Here two related identities share each row; each identity keeps its own delimiter pair.

LaTeX source

```
\begin{alignat*}{2}
 f(x)&=\left(x+1\right)^2
 &\qquad g(x)&=\left(x-1\right)^2\\
     &=x^2+2x+1
 &           &=x^2-2x+1
\end{alignat*}
```

Result

$$\begin{aligned} f(x) &= (x+1)^2 & g(x) &= (x-1)^2 \\ &= x^2 + 2x + 1 & &= x^2 - 2x + 1 \end{aligned}$$

3.5 Choose sensible break points

Break between complete terms or at a step in the calculation, where the reader can follow the continuation. Keep a fraction, an exponent or a matrix together. You choose the breaks; finaldelim arranges the resulting rows.

[5] Wikipedia, Normal distribution, estimation of parameters.

4 Use finaldelim in an existing document

4.1 Keep standard environment names, or opt in explicitly

By default, finaldelim extends `equation`, `align` and `alignat`, their starred forms and `\[...\]`. Existing displays can therefore use the new breaks without changing their environment names.

If you prefer to leave those standard interfaces alone, load the package with `override=false` and use its own environments:

LaTeX source

```
\begin{fdequation}[niji,render=draw]
 F&=\left[a+b+\\
  &\quad\frac{x}{y}+c\right]^2
\end{fdequation}
```

Result

$$F = \left[a + b + \frac{x}{y} + c\right]^2 \tag{5}$$

For this example, the preamble is

```
\usepackage[niji,render=draw,override=false]{finaldelim}
```

The fd-prefixed environments are always available, whether or not standard names are replaced. The option controls the display interfaces; delimiter sizing and niji still apply in native mathematics.

Explicit interface	Default standard name	Purpose
`fdequation`	`equation`	One number for the whole formula.
`fdequation*`	`equation*`, `\[...\]`	No automatic number.
`fdalign`, `fdalign*`	`align`, `align*`	Multiple rows, numbered or unnumbered.
`fdalignat`, `fdalignat*`	`alignat`, `alignat*`	Explicit column-pair count.
`fdsplit`	Not replaced by default	Embedded block, outer numbering.
`fdaligned`	Not replaced	Embedded block with vertical positioning.

`fdaligned` takes `[t]`, `[c]` or `[b]`, followed by an optional settings list. To align a block at its top, write:

```
\begin{fdaligned}[t][niji]
  ...
\end{fdaligned}
```

`fdsplit` takes an optional settings list.

For selective replacement, set `overrideequation`, `overridealign` or `overridealignat` in the package options. `overridesplit` is opt-in. These are preamble choices, not switches for individual formulas.

4.2 Already using physics?

Load `physics` and `finaldelim` in either order. Integration is automatic: delimiters made by `\qty`, `\abs`, `\norm` and bra–ket commands use the same sizing and niji colors.

```
\usepackage{physics}
\usepackage[niji]{finaldelim}
```

The earlier Gaussian relative-entropy expression can be written with physics commands.[6] The alignment markers and breaks are inside the argument of `\bqty`:

LaTeX source

```
\begin{equation}[niji]
 &D_{\mathrm{KL}}(\mathcal N_0
     \Vert\mathcal N_1)\\
 &=\frac12\bqty{
   \tr\qty(\Sigma_1^{-1}\Sigma_0)-k\\
 &\quad+(\mu_1-\mu_0)^{\mathsf T}
   \Sigma_1^{-1}(\mu_1-\mu_0)\\
 &\quad+\ln\frac{\det\Sigma_1}
                {\det\Sigma_0}}.
\end{equation}
```

Result

$$
\begin{aligned}
&D_{\mathrm{KL}}(\mathcal{N}_0 \| \mathcal{N}_1) \\
&= \frac{1}{2}\Bigg[\operatorname{tr}\left(\Sigma_1^{-1}\Sigma_0\right) - k \\
&\quad + (\mu_1 - \mu_0)^{\mathsf{T}} \Sigma_1^{-1} (\mu_1 - \mu_0) \\
&\quad + \ln \frac{\det \Sigma_1}{\det \Sigma_0} \Bigg].
\end{aligned}
\tag{6}
$$

Direct breaks are recognized in the automatic forms of `\qty`, `\quantity`, `\pqty`, `\bqty`, `\Bqty`, `\vqty`, `\abs`, `\absolutevalue` and `\norm`. Matrix commands such as `\mqty` and `\smqty` retain their internal rows and columns. Starred arguments, fractions, scripts and unknown macro arguments remain indivisible. Use explicit delimiters for outer breaks hidden inside an unsupported macro's argument.

[6] Wikipedia, Multivariate normal distribution, Kullback–Leibler divergence.

4.3 Inspect an unexpected alignment

If the alignment or row spacing is unclear, add `debug=all` to that display. The same formula below is first shown normally, then with its boxes and alignment points visible.

Normal output

LaTeX source

```
\begin{equation*}
 F&=\left[a+b+\\
  &\quad\frac{x}{y}+c\right]^2
\end{equation*}
```

Result

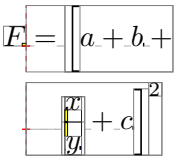
$$F = \left[a + b + \\ \quad \frac{x}{y} + c\right]^2$$

With diagnostics

LaTeX source

```
\begin{equation*}[debug=all]
 F&=\left[a+b+\\
  &\quad\frac{x}{y}+c\right]^2
\end{equation*}
```

Result



The alignment marks show where the rows meet at `&`. The box outlines show the space occupied by each row; the fraction makes the second row taller. The diagnostics are overlays, not additional mathematical content.

Use `debug=alignment` for alignment marks only, or `debug=boxes` for boxes only. Remove the option, or use `debug=none`, for the final document. Box outlines use `lua-visual-debug` and can be configured with its `\lvdset` interface.

5 Settings and installation reference

5.1 Set a default, then override it where needed

Supply settings when loading the package, change subsequent mathematics with `\FinalDelimSetup`, or supply local settings on a finaldelim display. The same display options work with either the standard or fd-prefixed names.

```
\usepackage[niji,render=draw]{finaldelim}
\FinalDelimSetup{scale=0.95}
\begin{fdequation}[niji=false,scale=0.9]
  \left[a+\frac{b}{c}\right]
\end{fdequation}
```

Sizing and rendering settings respect TeX groups. To limit any setup change to a passage, enclose that passage and its setup command in a group.

Option	Default	Effect
`sizing`	`weighted`	`native` restores the native size policy.
`lambda`	`20pt`	Physical decay length; smaller values emphasize nearby content.
`q`	`3`	Height emphasis; larger values favor tall content.
`scale`	`1`	Multiplier for the estimated radius.
`spanning`	`folded`	`folded` compresses middle rows for measurement; `full` retains their actual widths.
`fold-width`	`0.1em`	Width of a folded representative box; a positive TeX length.
`render`	`native`	`draw` enables continuous outlines.
`niji`	`false`	Nesting colors; `rainbow` is an alias.
`colors`	Built-in palette	A nonempty list of xcolor colors.
`debug`	`none`	`boxes`, `alignment`, or `all`.

`lambda` and `scale` must be positive; `q` must be nonnegative. The special value `q=0` selects the maximum-radius limit. Native sizing and explicitly sized delimiters are unaffected by the weighted size controls. Environment replacement options are listed in Section 4 and apply only when loading the package.

5.2 Install and compile

Install these nine files together beside your document or in a local TEXMF tree:

```
finaldelim.sty              finaldelim-luatex.sty
finaldelim-interface.sty    finaldelim-physics.sty
finaldelim.lua              finaldelim-size.lua
finaldelim-draw.lua         finaldelim-outline-fonts.lua
finaldelim-output.lua
```

Compile with LuaLaTeX. No Lua code is needed in the document. This development version does not yet provide tagged mathematical structure for its physical line-breaking backend.

5.3 DVI output

For a DVI workflow, compile with LuaLaTeX and convert with dvipdfmx:

```
lualatex --output-format=dvi document.tex
dvipdfmx document.dvi
```

Select the dvipdfmx color and graphics driver in the document, for example:

```
\documentclass[dvipdfmx]{article}
\usepackage{unicode-math}
\setmathfont{Latin Modern Math}
\usepackage[niji,render=draw]{finaldelim}
```

The package detects DVI mode and the graphics driver automatically. The same sizing, continuous drawing, colors, explicit line breaks, numbering and debug options apply. Direct PDF output remains available with the usual LuaLaTeX command and default document driver.

For PostScript or SVG, use `dvips` or `dvisvgm` as the document-class driver and convert the DVI with the matching command:

```
dvips -o document.ps document.dvi
dvisvgm --no-fonts --page=1- --bbox=papersize document.dvi
```

If driver detection needs an override, pass `dvi-driver=dvipdfmx`, `dvi-driver=dvips` or `dvi-driver=dvisvgm` when loading finaldelim. The default is `dvi-driver=auto`; this setting does not change the color or graphics package's driver.

dvisvgm uses its Ghostscript-backed PostScript handler for the drawn paths. Traditional dvips requires fonts it can read, such as TFM/Type 1 fonts; `\usepackage[OT1]{fontenc}` selects traditional text encoding in LuaLaTeX. For OpenType mathematics with `unicode-math`, use dvipdfmx or dvisvgm. The converter's font support is independent of finaldelim's drawn outlines. SVG has no device-dependent hairline, so zero-width debug strokes become explicit 0.1bp strokes there.

Maintained by Yu Zhai. LPPL 1.3c or later. Formula sources appear in footnotes; notation and breaks are adapted for the examples. The `lllhlll` profile and the short interface examples are schematic demonstrations.